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Workshop: 21st Century Raga: Digitizing North Indian Music

Abstract

This workshop describes methods for digitizing, analyzing, preserving and extending North Indian Classical Music. Custom built controllers, influenced by the Human Computer Interaction (HCI) Community, serve as new interfaces to gather musical gestures from a performing artist. Modified tabla, dholak, and sitar will be described. Experiments using Wearable sensors to capture ancillary gestures of a human performer will also be discussed. A brief history through the world of Musical Robotics will be followed by an introduction to the MahaDeviBot, a 12-armed solenoid-based drummer used to accompany a live sitar player. Presentation is full of video examples showing evolution of the body of work in the laboratory to the live performance stage. Live demonstrations will also be included in workshop..

Links to work:
http://www.digitalraja.com (video on bottom from Knowledge Network)

http://www.youtube.com/watch?v=I6hkLQVnCoQ



Biographical information

Ajay Kapur is the Director of the Interactive Music Technology, Intelligence and Design at California Institute of the Arts. He received an Interdisciplinary Ph.D. in 2007 from University of Victoria combining Computer Science, Electrical Engineering, Mechanical Engineering, Music and Psychology with a focus on Intelligence Music and Media Technology. Ajay graduated with a Bachelor in Science and Engineering Computer Science degree from Princeton University in 2002. He has been educated by music technology leaders including Dr. Perry R. Cook, Dr. George Tzanetakis, and Dr. Andrew Schloss, combined with mentorship from robotic musical instrument sculptors Eric Singer and the world famous Trimpin. A musician at heart, trained on Drumset, Tabla, Sitar and other percussion instruments from around the world, Ajay strives to push the technological barrier in order to make new music